

SECTION- A

A1

P. 20

ii) d) 55

Given,

x	-4	1	2	3
y	4	6	8	10

eqn of straight line,

$$y = ax + b \quad n = 4$$

normal eqn,

$$\sum y = a \sum x + bn \quad \text{--- (1)}$$

$$\sum xy = a \sum x^2 + b \sum x \quad \text{--- (2)}$$

x	y	x ²	xy
-4	4	16	-16
1	6	1	6
2	8	4	16
3	10	9	30

$$\sum x = 2 \quad \sum y = 28 \quad \sum x^2 = 30 \quad \sum xy = 36$$

$$\text{(1)} \Rightarrow 28 = a(2) + 4b \quad \text{--- (3)}$$

$$\text{(2)} \Rightarrow 36 = 30a + 2b \quad \text{--- (4)}$$

by solving (3) & (4)

$$a = 0.7586$$

$$b = 6.6206$$

$$\text{(*)} \Rightarrow y = 0.7586x + 6.6206$$

P. 20 eqn of straight line,

$$\text{normal eqn } y = ax + b \quad \text{--- (*)}$$

$$\sum y = a \sum x + bn \quad \text{--- (1)}$$

$$\sum xy = a \sum x^2 + b \sum x \quad \text{--- (2)}$$

x	y	x ²	xy
0	1	0	0
1	1.8	1	1.8
2	3.3	4	6.6
3	4.5	9	13.5
4	6.3	16	25.2
$\sum x = 10$	$\sum y = 16.9$	$\sum x^2 = 30$	$\sum xy = 47.1$

n = 4

$$\text{(1)} \Rightarrow 16.9 = 10a + 4b \quad \text{--- (3)}$$

$$\text{(2)} \Rightarrow 47.1 = 30a + 10b \quad \text{--- (4)}$$

by solving (3) & (4)

$$a = 0.97$$

$$b = 1.8$$

$$\text{(*)} \Rightarrow y = 0.97x + 1.8$$

A2

i) a) 6

ii) c) 38

Given,

$$iii) I_1 = [4, 1] A$$

$$I_2 = [-2, 2] A$$

$$I = [2, 8] A$$

W.K.T

$$I = aI_1 + bI_2$$

$$[2, 8] = a[4, 1] + b[-2, 2]$$

$$= [4a, a] + [-2b, 2b]$$

$$[2, 8] = [4a - 2b, a + 2b]$$

$$\Rightarrow 2 = 4a - 2b \quad \text{--- (1)}$$

$$8 = a + 2b \quad \text{--- (2)}$$

$$(1) + (2) \quad 2 = 4a - 2b$$

$$8 = a + 2b$$

$$10 = 5a$$

$$\boxed{a = 2}$$

$$(1) \Rightarrow 2 = 4(2) - 2b$$

$$2 = 8 - 2b$$

$$\Rightarrow 2b = 8 - 2 = 6$$

$$b = 6/2$$

$$\boxed{b = 3}$$

Sub $a=2, b=3$ in (2)

$$8 = 2 + 2(3) = 2 + 6$$

$$\Rightarrow 8 = 8$$

$\therefore$ It satisfy the linear combination.

iv) Let,

$$A = \begin{bmatrix} 0 & -2 & -2 \\ -1 & 1 & 2 \\ -1 & -1 & 2 \end{bmatrix}$$

chara eqn,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \quad \text{--- (1)}$$

$$S_1 = 0 + 1 + 2 = 3$$

$$S_2 = \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -2 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 0 & -2 \\ -1 & 1 \end{vmatrix}$$

$$= (2+2) + (0-2) + (0-2) = 0$$

$$S_3 = |A| = 0 + 2 \begin{vmatrix} -1 & 2 \\ -1 & 2 \end{vmatrix} - 2 \begin{vmatrix} -1 & 1 \\ -1 & -1 \end{vmatrix}$$

$$= 2(-2+2) - 2(1+1)$$

$$= 2(0) - 2(2) = -4$$

$$\boxed{S_3 = -4}$$

$$\Rightarrow \lambda^3 - 3\lambda^2 + 0\lambda + 4 = 0$$

$$\boxed{\lambda = -1, 2, 2}$$

are the eigen values.

$$P_1: \lambda = 2$$

$$[A - \lambda I] X = 0$$

$$\begin{bmatrix} 0-2 & -2 & -2 \\ -1 & 1-2 & 2 \\ -1 & -1 & 2-2 \end{bmatrix} X = 0$$

SECTION - B

$$\begin{bmatrix} -2 & -2 & -2 \\ -1 & -1 & 2 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \text{B1}$$

$$-2x_1 - 2x_2 - 2x_3 = 0 \quad \text{--- (1)}$$

$$-x_1 - x_2 + 2x_3 = 0 \quad \text{--- (2)}$$

$$-x_1 - x_2 + 0x_3 = 0 \quad \text{--- (3) iii)}$$

$$\begin{array}{ccccccc} -1 & -1 & 2 & -1 & -1 & & \\ & & & & & & \\ -1 & -1 & 0 & -1 & -1 & & \\ & & x_1 & & x_2 & & \end{array}$$

$$\frac{x_1}{0+2} = \frac{x_2}{-2+0} = \frac{x_3}{1-1}$$

$$\frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{0}$$

$$\therefore X = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}$$

$\therefore$ here 2 is larger (highest) value in this eigen vectors, the corresponding city is A $\Rightarrow$ city A has the highest flow value $\therefore$ city A is central hub or receives the highest traffic volume in the transportation network.

i) a) 1955

ii) a) upward

Given, $\sum x = 0$,
 $\sum x^2 = 28$, $\sum x^3 = 0$, $\sum x^4 = 196$,
 $\sum y = 6$, $\sum xy = 40$, $\sum x^2y = 6$

the quadratic eqn,

$$y = ax^2 + bx + c \quad \text{--- (*)}$$

w.k.t $n = 7$

normal equation,

$$\sum y = a \sum x^2 + b \sum x + cn \quad \text{--- (1)}$$

$$\sum xy = a \sum x^3 + b \sum x^2 + c \sum x \quad \text{--- (2)}$$

$$\sum x^2y = a \sum x^4 + b \sum x^3 + c \sum x^2 \quad \text{--- (3)}$$

$$\text{(1)} \Rightarrow 6 = 28a + 0b + 7c \quad \text{--- (4)}$$

$$\text{(2)} \Rightarrow 40 = 0a + 28b + 0c \quad \text{--- (5)}$$

$$\text{(3)} \Rightarrow 6 = 196a + 0b + 28c \quad \text{--- (6)}$$

From (4), (5), (6)

$$a = -0.2142$$

$$b = 1.4285$$

$$c = -1.7142$$

$$(*) y = -0.2142x^2 + 1.4285b + 1.7142c$$

9v) $P(x) = -2x^2 + 80x - 600$

$$a = -2,$$

$$b = 80, c = -600$$

$$\begin{aligned} &= -2(20)^2 + 80(20) - 600 \\ &= -800 + 1600 - 600 \\ &= 200 \end{aligned}$$

$$\text{Vertex } (h, k) = (20, 200)$$

a) x-Intercept,

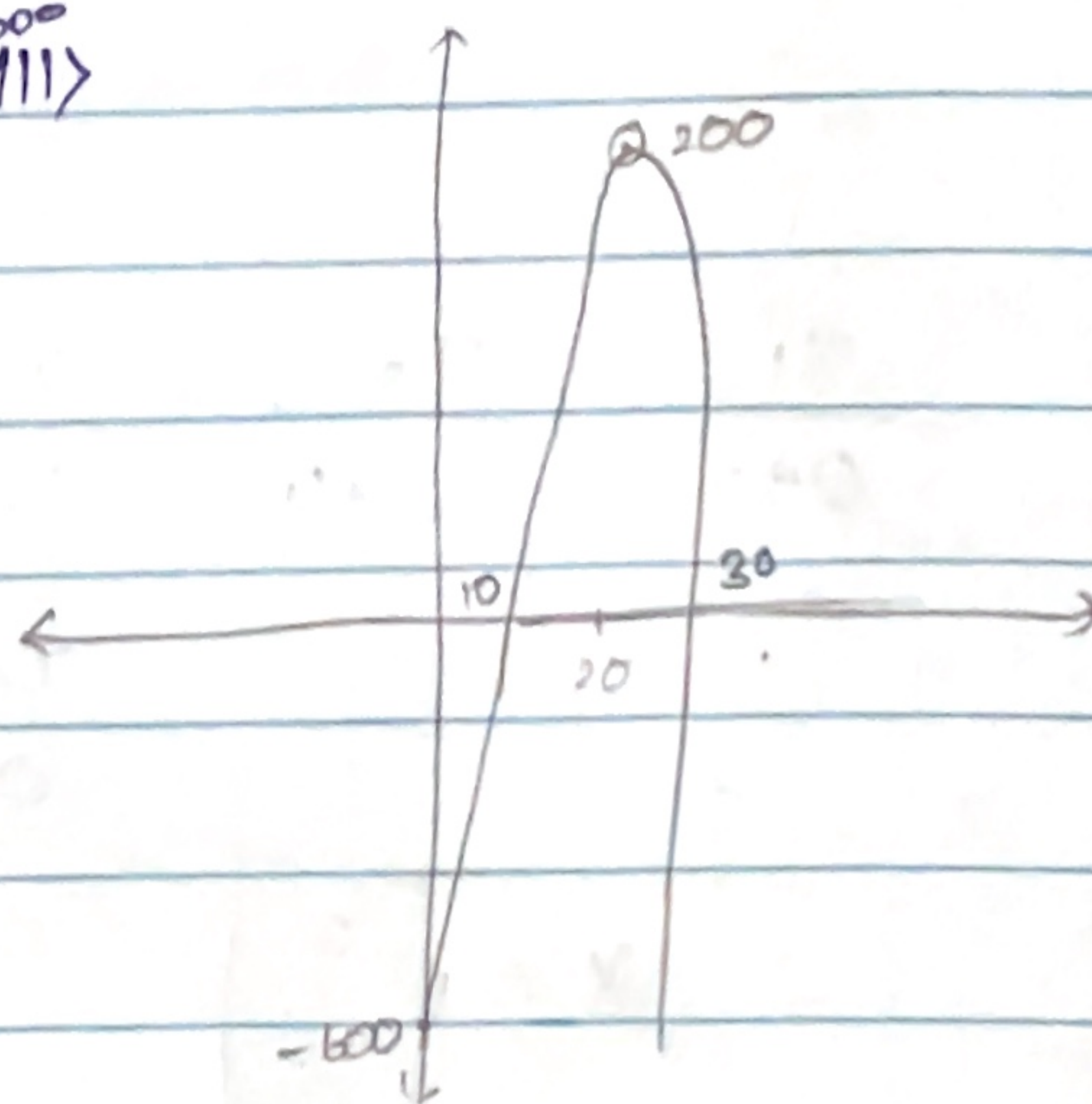
$$P(x) = y = 0$$

$$-2x^2 + 80x - 600 = 0$$

$$\boxed{x = 10, 30} \quad (x-10)(x-30) = 0$$

$$\boxed{x = 10} \quad \boxed{x = 30}$$

iii)



b) y-Intercept

$$x = 0$$

$$y = P(0) = -2(0)^2 + 80(0) - 600$$

$$\boxed{y = -600}$$

b) vertex (h, k)

$$h = -\frac{b}{2a} = \frac{-80}{2(-2)} = 20$$

$$\boxed{h = 20}$$

$$k = P(h) = P(20)$$

B3) i) d) Partial definite
ii) a) A^T

iii)

$$6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz = 0$$

$$A = \begin{bmatrix} \text{coeff } x^2 & \text{coeff } xy & \text{coeff } xz \\ \text{coeff } xy & \text{coeff } y^2 & \text{coeff } yz \\ \text{coeff } xz & \text{coeff } yz & \text{coeff } z^2 \end{bmatrix}$$

8.10
1.946

a) quadratic form to matrix

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

b) chara eqn,

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \quad (*)$$

$$S_1 = 6 + 3 + 3 = 12$$

$$S_2 = \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 6 & -2 \\ 2 & 3 \end{vmatrix}$$

$$= (9-1) + (18-4) + (18-4)$$

$$= 8 + 14 + 14 = 36$$

$$S_3 = |A|$$

$$= 6(9-1) + 2(-6+2) + 2(2-6)$$

$$= 6(8) + 2(-4) + 2(-2)$$

$$= 48 - 8 - 4 = 40 - 4$$

$$= 36$$

(*) $\Rightarrow$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 36 = 0$$

$$\lambda = 8, 1, 1$$

$$iv) A = \begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix} \quad \lambda^2 - S_1\lambda + S_2 = 0$$

a)

$$\lambda = -3 \quad X_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\lambda = 1$$

$$\lambda = 2$$

$$X_2 = ?$$

$$\begin{cases} S_1 = -1 \\ S_2 = -2 + 1 = 1 \end{cases}$$

$$[A - \lambda I]X = 0$$

$$\left(\begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -2-2 & 2-0 \\ 2-0 & 1-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-4x_1 + 2x_2 = 0 \quad \text{--- (1)}$$

$$2x_1 - x_2 = 0 \quad \text{--- (2)}$$

$$(1) \Rightarrow -4x_1 + 2x_2 = 0$$

$$(2) \times 2 \Rightarrow 4x_1 = 0$$

$\therefore A$ is symmetric matrix so their

eigen vectors are orthogonal to each other

$$X_1^T \cdot X_2 = 0$$

8.10.12
1.9463

$$[-2, 1] \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-2x_1 + x_2 = 0$$

SECTION - C

(C1)

i) c) 4

(C2)

i) c) a=2.5, b=10

ii)

x y x² x³ x⁴

0 1 0 0 0

1 1.8 1 1 1

2 1.3 4 8 16

3 2.3 9 27 81

4 6.3 16 64 256

36.16
96
256

x⁵ xy x²y x³y

0 0 0 0

1 1.8 1.8 1.8

2 2.6 5.2 10.4

3 6.9 20.7 62.1

4 25.2 100.8 403.2

36.5

$$\sum x = 10$$

$$\sum y = 12.7$$

$$\sum x^2 = 30$$

$$\sum x^3 = 100$$

x⁶

0

1

64

729

4096

$$\Rightarrow 12.7 = 4a + 10b + 30c + 100d$$

$$36.5 = 10a + 30b + 100c + 354d$$

$$128.5 \quad \cancel{477.5} = 30a + 100b + 354c + 1300d$$

$$477.5 = 100a + 354b + 1300c + 4890d.$$